

Weekly Progress Report

AgAI Pesticide Residue & Leaf Classification Project

Week 2: 3/2/2026

Overview

This week I met with Anthony Bilic to discuss modeling strategy and the limitations of the current dataset. From that discussion, we defined the initial deliverables for the modeling pipeline and identified a baseline approach for working with the limited ICP-validated samples.

Meetings and Technical Alignment

- Met with Anthony Bilic to discuss how to approach the limited dataset and structure the modeling pipeline.
- Key takeaway: begin with simple baseline models before extending the system toward the more complex goals described in the project proposal.
- Given the limited number of ICP-validated samples and overall dataset size, we discussed starting with a **threshold-based binary classification model** to determine whether pesticide levels are sufficient or require reapplication. 
 - This aligns closely with the biological objective of the project.
 - It also provides a practical starting point given the current dataset constraints.
 - Anthony also recommended building a **clean baseline pipeline** rather than immediately extending last year's implementation.
 - This helps ensure preprocessing, dataset structure, and evaluation procedures are fully understood before integrating existing code.

Baseline Modeling Approach

- I plan to begin with two baseline models to evaluate what signal exists in the current dataset.
- Baseline A – Binary Pesticide Decision
 - Model: CNN → MLP → binary output
 - Objective: predict whether pesticide levels are above or below the reapplication threshold.
 - Loss: BCEWithLogitsLoss
 - Purpose: establish a measurable baseline for pesticide-related prediction.
- Baseline B – Residue segmentation (if masks are available)
 - Model: U-Net segmentation architecture
 - Objective: predict residue masks at the pixel level.
 - Potential benefit: segmentation may provide features such as residue coverage or spatial distribution that correlate with pesticide levels better than image-level classification.
 - If segmentation masks exist, this could serve as a stronger physical proxy for pesticide coverage.
- The goal of these baselines is to quickly evaluate the available signal in the data before exploring more complex architectures such as transformer-based approaches.

Coordination with Previous Work

- Currently coordinating with William Bu to obtain access to:
 - last year's code repository
 - the Roboflow dataset used for training and annotation
- Plan for reviewing last year's implementation:
 - Inspect preprocessing steps
 - Run the pipeline locally
 - Compare dataset splits and training procedures
- Once the pipeline is understood, I will also explore extending the previous model to support **multi-label classification** for the leaf classification task.

Dataset Assessment and Collaboration with the Chemistry Team

- Building a clean pipeline will also help clarify what dataset structure or metadata may be needed from the data collection side.
- Potential dataset considerations include:
 - consistent image framing and lighting
 - plant identifiers for dataset splitting
 - metadata needed to prevent leakage between train/test sets
- This information will allow me to coordinate more effectively with Giulio and Melissa to ensure the dataset supports the modeling objectives.

Plan for the Upcoming Week

1. Clarify dataset assets and ground truth
 - Confirm the exact labels currently available (e.g., condition labels; clean, canker, dirt).
 - Determine whether segmentation masks or residue annotations exist in the Roboflow dataset.
 - Confirm how pesticide levels are labeled:
 - plant-level vs leaf-level
 - ICP vs estimated.
2. Begin building a baseline training pipeline
 - Implement a dataloader and preprocessing pipeline.
 - Standardize image orientation and resizing.
 - Implement training and evaluation loops.
 - Implement **plant-level dataset splits** to prevent leakage between training and test data.
3. Implement Baseline A (binary pesticide decision)
 - CNN-based classifier predicting whether pesticide levels are above or below the threshold.
4. Investigate Baseline B (segmentation)
 - Inspect Roboflow annotations to determine whether segmentation masks exist.
 - If masks exist, begin implementing a U-Net segmentation baseline.

5. Inspect and run William's previous implementation

- Run the existing pipeline locally.
- Compare preprocessing and dataset handling approaches.
- Begin planning the upgrade to multi-label classification once the codebase is understood.

Overall goal

The immediate focus is to establish simple, interpretable baselines and gain a clearer understanding of the dataset. This will make it easier to extend the system toward multi-label leaf classification and pesticide residue prediction once the underlying pipeline is stable.